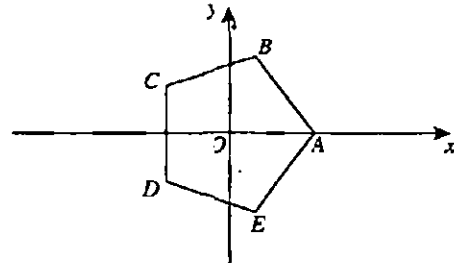


QUESTION 1	MARKS	QUESTION 2	BEGIN A NEW PAGE	MARKS
(a) Find $\int \frac{dx}{\sqrt{9-4x^2}}$.	1	(a) Let $z = \frac{2-4i}{1+i}$.		
		(i) Find $\bar{z}$, giving your answer in the form $a+bi$, where a and b are real.		2
(b) (i) Find real constants A , B and C such that	3	(ii) Find iz .		1
$\frac{x^2+5x+2}{(x^2+1)(x+1)} = \frac{Ax+B}{x^2+1} + \frac{C}{x+1}.$				
(ii) Hence find $\int \frac{x^2+5x+2}{(x^2+1)(x+1)} dx$.	2	(b) Find a and b if $(a+ib)^2 = 3-4i$, where a and b are real and $e > 0$.		2
(c) Evaluate $\int_1^5 x\sqrt{2x-1} dx$.	3	(c) Consider the region defined by $ z-4i \leq 3$.		
		(i) Sketch the region.		1
(d) Evaluate $\int_0^1 x^5 e^{x^3} dx$.	3	(ii) Determine the maximum value of $ z $.		1
		(iii) Determine the maximum value of $\arg z$, where $-\pi < \arg z \leq \pi$.		2
(e) (i) Simplify $\sin(A-B) + \sin(A+B)$.	1	(d)		
(ii) Hence find $\int \sin 5x \cos 3x dx$.	2			
		In the diagram above, the complex numbers z_0, z_1, z_2, z_3, z_4 are represented by the vertices of a regular polygon with centre O and vertices A, B, C, D, E respectively.		
		Given that $z_0 = 2$:		
		(i) Express z_2 in modulus-argument form.		2
		(ii) Find the value of z_2^5 .		2
		(iii) Show that the perimeter of the pentagon is $20\sin\frac{\pi}{5}$.		2

QUESTION 3 BEGIN A NEW PAGE

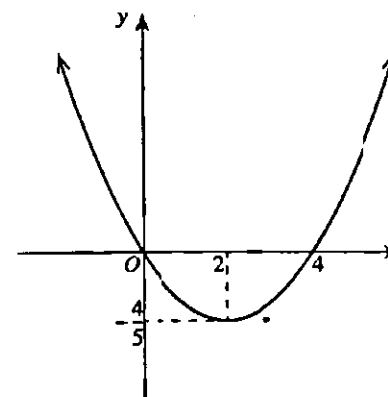
MARKS

- (a) Let α , β and γ be the roots of $x^3 - 7x^2 + 18x - 7 = 0$
- (i) Find a cubic equation that has roots, $1 + \alpha^2$, $1 + \beta^2$ and $1 + \gamma^2$. 2
- (ii) Hence or otherwise, find the value of $(1 + \alpha^2)(1 + \beta^2)(1 + \gamma^2)$. 1
- (b) (i) The polynomial equation $p(x) = 0$ has a root α of multiplicity 3. Show that α is a root of $p'(x) = 0$ and is of multiplicity 2. 2
- (ii) The polynomial $q(x) = x^6 + ax^5 + bx^4 - x^2 - 2x - 1$ has a quadratic factor of $x^2 + 2x + 1$. Find a and b . 2
- (iii) Consider the polynomial $r'(x) = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}$ where $r(0) = 1$. 3
- Show that $r(x)$ has no double roots.
- (c) The acceleration of a particle moving in a straight line, starting from a position 2 metres on the positive side of the origin, with a velocity of 1.5 ms^{-1} is given by $\frac{dv}{dt} = \frac{9 - x^2}{x^4}$.
- (i) Show that the velocity in ms^{-1} of the particle can be expressed as $v = \frac{\sqrt{2x(x^3 + x^2 - 3)}}{x^2}$. 3
- (ii) Describe the behaviour of the velocity of the particle after it passes $x = 3$. 2

QUESTION 4 BEGIN A NEW PAGE

MARKS

(a)



The sketch above shows the parabolic curve $y = f(x)$ where

$$f(x) = \frac{x^2 - 4x}{5}.$$

Without the use of calculus, draw sketches of the following, showing intercepts, asymptotes and turning points:

- (i) $y = |f(x)|$, 1
- (ii) $y = \frac{1}{f(x)}$, 2
- (iii) $y = \frac{x}{5}|x - 4|$, 2
- (iv) $y = \tan^{-1}(f(x))$, 2

Question 4 continued on page 5

- (b) The circle $x^2 + y^2 = 4$ is revolved about the y -axis to generate a solid sphere. A cylindrical hole whose diameter is 2 units and whose axis is the y -axis is then removed from the sphere, leaving a solid S .

Figure 1 below shows a three-dimensional perspective and Figure 2 shows a cross-sectional view.

Using the method of cylindrical shells, find the volume of S .

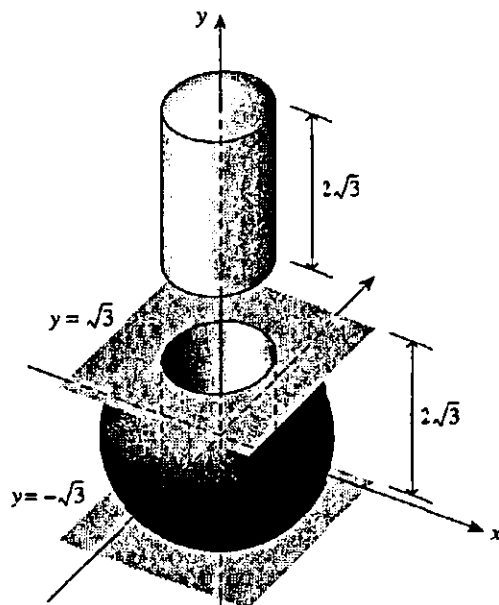


Figure 1

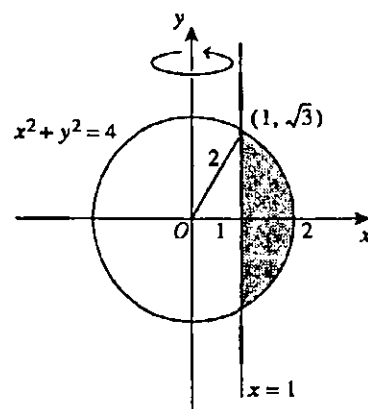


Figure 2

- (c) The length of an arc joining $P(a, c)$ and $Q(b, d)$ on a smooth, continuous curve $y = f(x)$ is given by

$$\text{arc length} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

Consider the curve defined by $y = \frac{x^2}{4} - \frac{\ln x}{2}$.

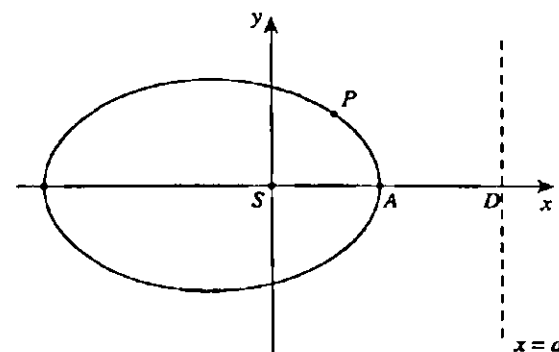
- (i) Show that $1 + \left(\frac{dy}{dx}\right)^2 = \frac{1}{4}\left(x + \frac{1}{x}\right)^2$.

- (ii) Find the length of the arc between $x = 1$ and $x = e$.

End of Question 4

page 5 of 11

4 (a)



Consider the ellipse sketched above of eccentricity e with one focus S at the origin and its corresponding directrix at $x = d$.

- (i) If P corresponds to the complex number z , where $z = r(\cos \theta + i \sin \theta)$, use the focus-directrix definition of an ellipse to show that the ellipse can be expressed as

$$r = \frac{ed}{1 + e \cos \theta}.$$

- (ii) Hence draw the ellipse represented by

$$r = \frac{33}{5 + 3 \cos \theta}$$

showing the coordinates of the points A and D .

[There is no need to find the coordinates of any other point, or to write the Cartesian equation of the ellipse.]

- (b) Consider the curve $x^2 - xy + y^2 = 3$.

- (i) Show that $\frac{dy}{dx} = \frac{2x-y}{x-2y}$.

- (ii) Hence find the two stationary points on the curve.

- (iii) Find the values of x where there are vertical tangents.

Question 5 continued on page 7

page 6 of 11

QUESTION 5 CONTINUED

MARKS

(c) Consider the complex number $z = \cos \theta + i \sin \theta$.

(i) Using de Moivre's theorem, show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$, for any integer n . 1

(ii) Hence or otherwise express $\left(z + \frac{1}{z}\right)^6$ in the form $A \cos 6\theta + B \cos 4\theta + C \cos 2\theta + D$, 2
where A, B, C and D are real constants.

(iii) Hence evaluate $\int_0^{\frac{\pi}{4}} \cos^6 \theta d\theta$. 2

End of Question 5

QUESTION 6

BEGIN A NEW PAGE

MARKS

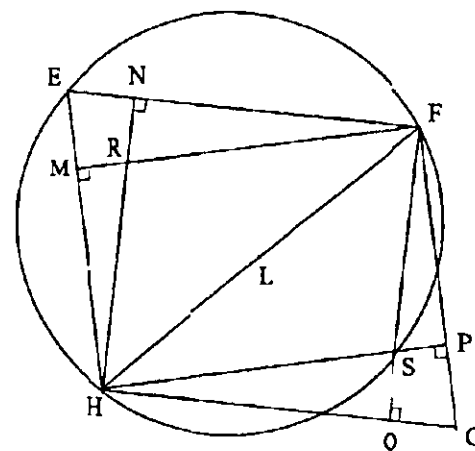
(a) (i) On the same set of axes, sketch the graphs of 3
 $y = \cos^{-1}\left(\frac{x-2}{2}\right)$ and $y = \frac{\pi}{2} + \tan^{-1}(x-2)$

(ii) From your graph, or otherwise, solve the inequality 1
 $\cos^{-1}\left(\frac{x-2}{2}\right) - \tan^{-1}(x-2) \leq \frac{\pi}{2}$

(b) Find the smallest positive integer p such that 4

$$\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^p = \frac{1}{2}(-1+i\sqrt{3})$$

(c) The vertices E, F and H of the parallelogram $EFGH$ lie on the circle. L is the midpoint of the diagonal FH .
 R is the point of intersection of the perpendicular heights EN and FM of the triangle EFH .
 S is the point of intersection of the perpendicular heights HP and FQ of the triangle FGH .

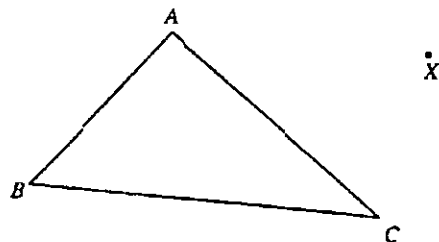


i) Prove that point S lies on the circle. 2

ii) Prove that the points R, L and S are collinear. 4

iii) Show that the hexagon $MNFPQH$ is cyclic. 1

(a)

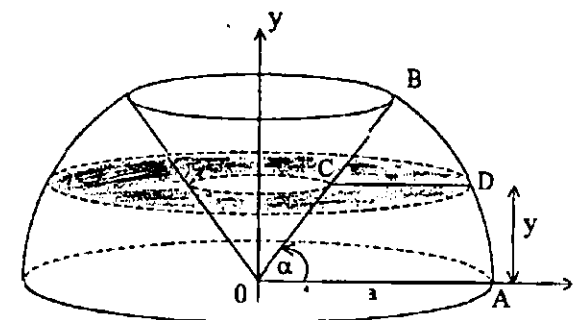


The diagram above shows a point X outside a triangle ABC .

Show that $AX + BX + CX > \frac{AB + BC + CA}{2}$.

2

- (c) The sector OAB with an angle α at the center $(0,0)$ is rotated about the y axis to form a solid. When the sector is rotated, the line segment CD at height y sweeps out an annulus as shown in the diagram below.



- (i) Show that the area of the annulus is $\pi(a^2 - y^2 \csc^2 \alpha)$.

2

- (ii) Find the volume of the solid.

3

End of Question 7

Question 7 continued on page 10

(a) Let x, y, z and w be positive real numbers.

(i) Prove that $\frac{x}{y} + \frac{y}{x} \geq 2$.

2

(ii) Deduce that $\frac{x+y+z}{v} + \frac{w+y+z}{x} + \frac{w+x+z}{y} + \frac{v+x+y}{z} \geq 12$.

2

(iii) Hence prove that if $x + y + z + w = 1$, then $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} + \frac{1}{w} \geq 16$.

2

(b) Let $J_n = \int_0^1 x^n e^{-x} dx$, where $n \geq 0$.

(i) Show that $J_0 = 1 - \frac{1}{e}$.

1

(ii) Show that $J_n = nJ_{n-1} - \frac{1}{e}$, for $n \geq 1$.

2

(iii) Show that $J_n \rightarrow 0$ as $n \rightarrow \infty$.

1

(iv) Deduce by the principle of mathematical induction that for all $n \geq 0$,

4

$$J_n = n! - \frac{n!}{e} \sum_{r=0}^n \frac{1}{r!}.$$

(v) Conclude that $e = \lim_{n \rightarrow \infty} \left(\sum_{r=0}^n \frac{1}{r!} \right)$.

1

End of paper

$$\begin{aligned} \text{a) } \int \frac{dx}{\sqrt{9-4x^2}} &= \frac{1}{2} \int \frac{dx}{\sqrt{\left(\frac{3}{2}\right)^2 - x^2}} \\ &= \frac{1}{2} \sin^{-1} \frac{2x}{3} + C \end{aligned} \quad [1]$$

$$\begin{aligned} \text{b) } \frac{x^2+5x+2}{(x^2+1)(x+1)} &= \frac{Ax+B}{x^2+1} + \frac{C}{x+1} \\ &= \frac{(Ax+B)(x+1) + C(x^2+1)}{(x^2+1)(x+1)} \end{aligned}$$

$$\therefore (Ax+B)(x+1) + C(x^2+1) = x^2+5x+2 \quad (1) \quad [1]$$

$$\begin{aligned} \text{Put } x &= -1 & 2C &= -2 \\ & & C &= -1 \end{aligned}$$

$$\begin{aligned} \text{Put } x &= 0 & B+C &= 2 \\ & & \therefore B &= 3 \end{aligned}$$

$$\begin{aligned} \text{Equate coeff of } x^2 & \\ A+C &= 1 \\ \therefore A &= 2 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} -1 \text{ for} \\ \text{each} \\ \text{mistake} \end{array} \quad [2]$$

$$(ii) \therefore \int \frac{x^2+5x+2}{(x^2+1)(x+1)} dx = \int \frac{2x+3}{x^2+1} - \frac{1}{x+1} dx$$

$$= \int \frac{2x dx}{x^2+1} + \int \frac{3}{x^2+1} dx - \int \frac{1}{x+1} dx \quad [1]$$

$$= \ln(x^2+1) + 3 \tan^{-1} x - \ln(x+1) + C$$

$$= \ln \frac{x^2+1}{x+1} + 3 \tan^{-1} x + C \quad [1]$$

$$(c) \text{ Put } u = \sqrt{2x-1}$$

$$\therefore u^2 = 2x-1$$

$$dx = u du$$

$$\text{when } x=5 \quad u=3$$

$$\text{when } x=1 \quad u=1$$

$$\therefore \int_1^5 x \sqrt{2x-1} dx = \int_1^3 \frac{u^2+1}{2} \cdot u \cdot u du \quad [1]$$

$$= \int_1^3 \frac{u^4+u^2}{2} du$$

$$= \left[\frac{u^5}{10} + \frac{u^3}{6} \right]_1^3 \quad [1]$$

$$= \frac{3^5}{10} + \frac{3^3}{6} - \frac{1}{10} - \frac{1}{6}$$

$$= \frac{428}{15} \text{ or } 28\frac{8}{15} \quad [1]$$

$$d) \int_0^1 x^5 e^{x^3} dx$$

$$= \int_0^1 x^3 d\left(\frac{e^{x^3}}{3}\right) \quad [1]$$

$$= \left[\frac{1}{3} x^3 e^{x^3} \right]_0^1 - \frac{1}{3} \int_0^1 e^{x^3} d(x^3) \quad [1]$$

$$= \frac{1}{3} e - \int_0^1 x^2 e^{x^3} dx$$

$$= \frac{1}{3} e - \left[\frac{1}{3} e^{x^3} \right]_0^1$$

$$= \frac{1}{3} e - \left(\frac{1}{3} e - \frac{1}{3} \right)$$

$$= \frac{1}{3} \quad [1]$$

$$e) (i) \sin(A+B) + \sin(A-B)$$

$$= 2 \sin A \cos B \quad [1]$$

$$(ii) \int \sin 5x \cos 3x dx$$

$$= \frac{1}{2} \int (\sin 8x + \sin 2x) dx \quad [1]$$

$$= -\frac{1}{16} \cos 8x - \frac{1}{4} \cos 2x + C \quad [1]$$

Q2

$$a) z = \frac{2-4i}{1+i}$$

$$(i) z = \frac{2-4i}{1+i} \times \frac{1-i}{1-i} = \frac{(2-4i)-6i}{2}$$

$$= -1-3i$$

$$\therefore \bar{z} = -1+3i$$

$$(ii) iz = i(-1-3i) = 3-i$$

$$(b) (a+ib)^2 = 3-4i$$

$$a^2-b^2+2abi = 3-4i$$

$$\therefore a^2-b^2 = 3 \quad (1)$$

$$2ab = -4$$

$$\therefore b = -\frac{2}{a} \quad (2)$$

Put (2) into (1)

$$a^2 - \frac{4}{a^2} = 3$$

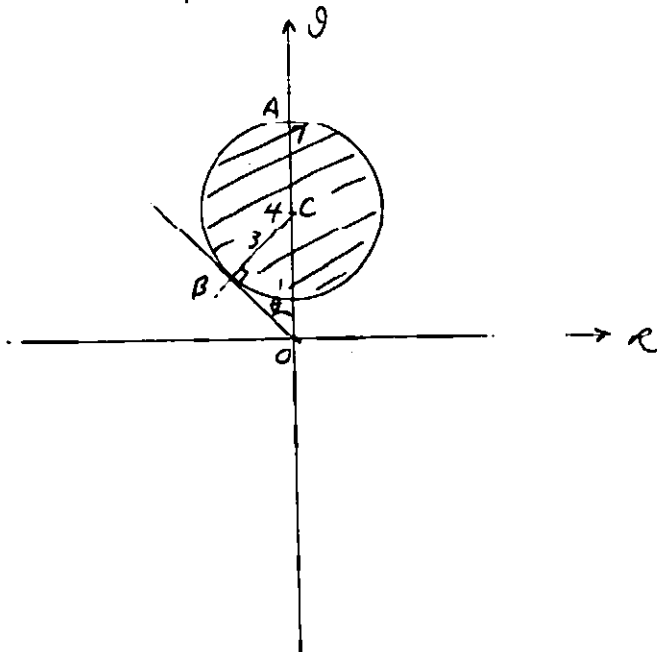
$$\therefore a^4 - 3a^2 - 4 = 0$$

$$(a^2+1)(a^2-4) = 0$$

$$\therefore a = 2 \quad \text{since } a > 0$$

$$b = -1$$

c) (i)



(ii) MAX $|z|$ occurs at A

$$\max |z| = 7$$

(iii) In $\triangle OBC$

$$\sin \theta = \frac{3}{4}$$

$$\therefore \theta = \sin^{-1}\left(\frac{3}{4}\right)$$

$$\therefore \max \text{ value of } \arg z = \frac{\pi}{2} + \sin^{-1}\left(\frac{3}{4}\right)$$

$$d) (i) \angle COA = \frac{2\pi}{5}$$

$$\therefore z_2 = 2 \operatorname{cis} \frac{2\pi}{5}$$

$$(ii) z_2^5 = 2^5 (\operatorname{cis} \frac{2\pi}{5})^5$$

$$= 2^5 \operatorname{cis} 2\pi$$

$$= 2^5$$

$$= 32$$

(iii) $\angle AOM$

$$= \frac{1}{2} \angle AOB$$

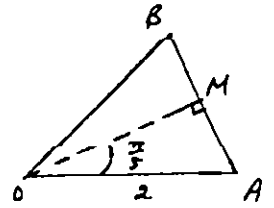
$$= \frac{\pi}{5}$$

$$AM = 2 \sin \frac{\pi}{5}$$

$\therefore$ Perimeter

$$= 10 AM$$

$$= 20 \sin \frac{\pi}{5}$$



Q3. (a) (i) Put $y = 1+x^2 \therefore x = \sqrt{y-1}$

$\therefore$ The transformed equation is

$$(\sqrt{y-1})^3 - 7(y-1) + 18\sqrt{y-1} - 7 = 0$$

$$\sqrt{y-1}(y-1+18) = 7(y-1+1)$$

$$\sqrt{y-1}(y+17) = 7y$$

$$\therefore (y-1)(y+17)^2 = 49y^2$$

$$(y-1)(y^2+34y+289) = 49y^2$$

$$y^3 - 16y^2 + 255y - 289 = 0$$

$$x^3 - 16x^2 + 255x - 289 = 0 \quad (1)$$

(ii) $(1+\alpha^2x)(1+\beta^2x)(1+\gamma^2x)$

= product of roots of (1)

$$= 289$$

b(i) Let $P(x) = (x-\alpha)^3 Q(x)$ where $Q(x) \neq 0$

$$\therefore P'(x) = 3(x-\alpha)^2 Q(x) + (x-\alpha)^3 Q'(x)$$

$$= (x-\alpha)^2 [3Q(x) + (x-\alpha)Q'(x)]$$

$$\therefore x=\alpha \text{ is a double root of } P(x)$$

(ii) $x^2 + 2x + 1 = (x+1)^2$

$\therefore x = -1$ is a double root of $q(x)$.

Hence by (i) $x = -1$ is a root of $q'(x)$.

$$\therefore q(-1) = q'(-1) = 0$$

$$1 - a + b - 1 + 2 - 1 = 0$$

$$a - b = 1 \quad (1)$$

$$-6 + 5a - 4b + 2 - 2 = 0$$

$$5a - 4b = 6 \quad (2)$$

$$4a - 4b = 4 \quad (3)$$

$$(1) \times 4$$

$$(2) - (3)$$

put into (1)

$$a = 2$$

$$b = 1$$

(iii) $r'(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$

$$r(x) = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots + \frac{x^{m+1}}{(m+1)!} + C$$

$$\therefore r(0) = 1$$

$$\therefore 1 = C$$

$$\therefore r(x) = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!}$$

suppose $x = \alpha$ is a double root of $r(x) = 0$

$$\text{then } r(\alpha) = r'(\alpha) = 0$$

ie $1 + \alpha + \frac{\alpha^2}{2!} + \frac{\alpha^3}{3!} + \dots + \frac{\alpha^n}{n!} = 0 \quad (1)$

$$1 + \alpha + \frac{\alpha^2}{2!} + \frac{\alpha^3}{3!} + \dots + \frac{\alpha^n}{n!} + \frac{\alpha^{n+1}}{(n+1)!} = 0 \quad (2)$$

$$(2) - (1) \quad \frac{\alpha^{n+1}}{(n+1)!} = 0$$

$$\therefore \alpha = 0$$

$$\text{but } r(0) = 1 \text{ and } r'(0) = 1$$

$\therefore \alpha = 0$ is not a double root

Hence $r(x) = 0$ does not have any double root.

c(i) $\frac{dv}{dt} = \frac{d}{dx}\left(\frac{v^2}{2}\right) = \frac{9-x^2}{x^4}$

$$\therefore \int d\left(\frac{v^2}{2}\right) = \int \left(\frac{9}{x^4} - \frac{1}{x^2}\right) dx$$

$$\frac{v^2}{2} = -\frac{3}{x^3} + \frac{1}{x} + C$$

since $v = 1.5$ when $x = 2$

$$\frac{1.5^2}{2} = -\frac{3}{2^3} + \frac{1}{2} + C$$

$$\therefore C = 1$$

$$\therefore v^2 = 2\left(1 + \frac{1}{x} - \frac{3}{x^3}\right)$$

$$= \frac{2(x^4 + x^3 - 3x)}{x^4}$$

$$\therefore v = \frac{\sqrt{2x(x^3 + x^2 - 3)}}{x^2}$$

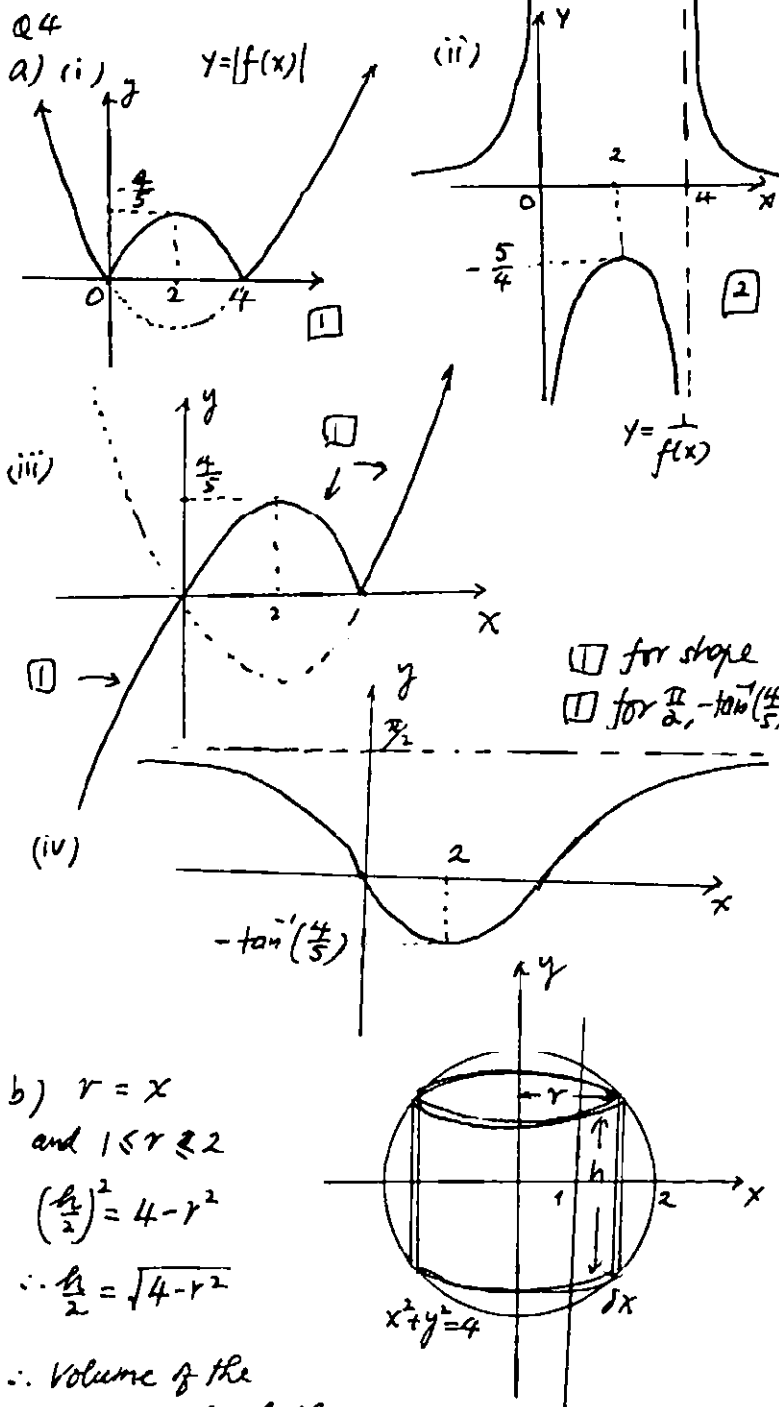
(iii) $a = \frac{9-x^2}{x^4}$

$\therefore a < 0$ for all $x < -3$ or $x > 3$

$\therefore$ The particle will slow down after it passes $x = 3$ towards the right.

$$\text{and } \lim_{x \rightarrow \infty} v = \lim_{x \rightarrow \infty} \sqrt{\frac{2(x^4 + x^3 - 3x)}{x^4}} = \sqrt{2}$$

$\therefore$ The speed of the particle will eventually be $\sqrt{2} \text{ ms}^{-1}$, ie it will never stop.



$\therefore$ Volume of the cylindrical shell

$$\delta V = 2\pi r h \delta x$$

$$= 4\pi x \sqrt{4 - x^2} \delta x \quad \square$$

$$\therefore \text{Volume of } S = 4\pi \int_1^2 x \sqrt{4 - x^2} dx \quad \square$$

Put $u = 4 - x^2 \quad \therefore du = -2x dx$

When $x = 2, u = 0$

When $x = 1, u = 3$

$$\therefore V = -\frac{4\pi}{2} \int_3^0 \sqrt{u} du \quad \square$$

$$= \left(\frac{2}{3}\right) 2\pi \left[u^{\frac{3}{2}} \right]_0^3$$

$$= 4\pi \sqrt{3} \text{ unit}^3 \quad \square$$

(c) $y = \frac{x^2}{4} \ln \frac{x}{2}$

(i) $\frac{dy}{dx} = \frac{x}{2} - \frac{1}{2x} \quad \square$

$$\therefore 1 + \left(\frac{dy}{dx}\right)^2 = 1 + \left(\frac{x}{2} - \frac{1}{2x}\right)^2$$

$$= 1 + \frac{x^2}{4} - \frac{1}{2} + \frac{1}{4x^2}$$

$$= \frac{x^2}{4} + \frac{1}{2} + \frac{1}{4x^2}$$

$$= \frac{1}{4} \left(x^2 + 2 + \frac{1}{x^2}\right)$$

$$= \frac{1}{4} \left(x + \frac{1}{x}\right)^2 \quad \square$$

(ii) Arc length

$$= \int_1^e \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int_1^e \sqrt{\frac{1}{4} \left(x + \frac{1}{x}\right)^2} dx$$

$$= \int_1^e \frac{1}{2} \left(x + \frac{1}{x}\right) dx \quad \square$$

$$= \frac{1}{2} \left[\frac{x^2}{2} + \ln x \right]_1^e$$

$$= \frac{1}{2} \left(\frac{e^2}{2} + \ln e - \frac{1}{2} \right)$$

$$= \frac{1}{4} (e^2 + 1) \text{ units} \quad \square$$

Q5

(a)(i) Coordinates of P is $(r \cos \theta, r \sin \theta)$

$\therefore$ $\perp$ distance from P to directrix $x = d$ is

$$PN = d - r \cos \theta$$

$$PS = r$$

$$\frac{PS}{PN} = e$$

$$\therefore PS = ePN$$

$$r = e(d - r \cos \theta)$$

$$= ed - er \cos \theta$$

$$r(1 + e \cos \theta) = ed$$

$$\therefore r = \frac{ed}{1 + e \cos \theta}$$

$$(ii) \text{ By comparing } r = \frac{33}{5 + 3 \cos \theta} = \frac{33/5}{1 + \frac{3}{5} \cos \theta} \quad \square$$

$$\therefore r = \frac{ed}{1 + e \cos \theta}$$

$$ed = \frac{33}{5}$$

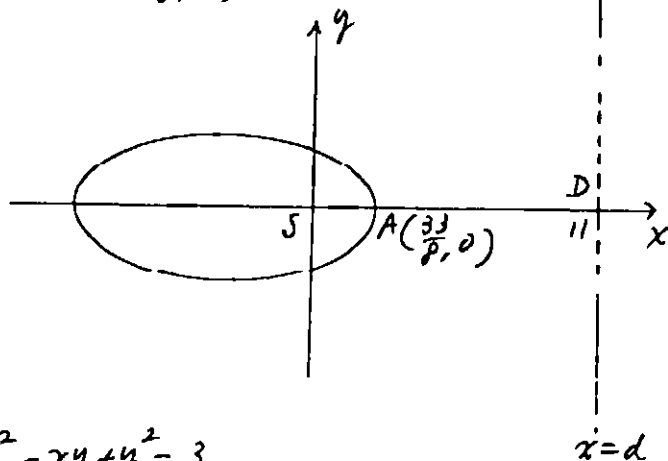
$$e = \frac{3}{5}$$

$$\therefore d = 11$$

$\therefore$ At A on the ellipse, $\theta = 0$

$$r = \frac{33}{5 + 3} = \frac{33}{8}$$

$$\therefore A \text{ is } \left(\frac{33}{8}, 0\right)$$



$$b) (i) x^2 - xy + y^2 = 3$$

$$2x - y - x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0$$

$$2x - y = (x - 2y) \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{2x - y}{x - 2y}$$

(ii) At stationary points, $\frac{dy}{dx} = 0$

$$\therefore 2x - y = 0$$

$$y = 2x$$

(1) $\square$

$$\text{Put } y = 2x \text{ into } x^2 - xy + y^2 = 3 \quad (2)$$

$$x^2 - x(2x) + (2x)^2 = 3$$

$$x^2(1 - 2 + 4) = 3$$

$$x^2 = 1$$

$$x = \pm 1$$

$\therefore$ the two stationary pts are

$$(1, 2) \text{ and } (-1, -2) \quad \square$$

(iii) When the tangent is vertical,

$$x - 2y = 0$$

$$x = 2y$$

$$\text{Put } y = \frac{x}{2} \text{ into (2)}$$

$$x^2 - x\left(\frac{x}{2}\right) + \left(\frac{x}{2}\right)^2 = 3$$

$$4x^2 - 2x^2 + x^2 = 12$$

$$x^2 = 4$$

$$x = \pm 2 \quad \square$$

$$c) (i) z^n = (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

$$\frac{1}{z^n} = (\cos \theta + i \sin \theta)^{-n} = \cos n\theta - i \sin n\theta$$

$$\therefore z^n + \frac{1}{z^n} = 2 \cos n\theta \quad \square$$

$$(ii) \left(z + \frac{1}{z}\right)^6 = z^6 + 6z^4 + 15z^2 + 20 + \frac{15}{z^2} + \frac{6}{z^4} + \frac{1}{z^6}$$

$$= \left(z^6 + \frac{1}{z^6}\right) + 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) + 20$$

$$= 2 \cos 6\theta + 12 \cos 4\theta + 30 \cos 2\theta + 20 \quad \square$$

$$(iii) \text{ From (ii) } 64 \cos^6 \theta = 2 \cos 6\theta + 12 \cos 4\theta + 30 \cos 2\theta + 20$$

$$\therefore \int_0^{\frac{\pi}{2}} \cos^6 \theta d\theta = \frac{1}{32} \int_0^{\frac{\pi}{2}} (2 \cos 6\theta + 12 \cos 4\theta + 30 \cos 2\theta + 20) d\theta$$

$$= \frac{1}{32} \left[\frac{2 \sin 6\theta}{6} + \frac{12 \sin 4\theta}{4} + \frac{30 \sin 2\theta}{2} + 20\theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{32} \left[6 \sin \frac{3\pi}{2} + 3 \sin \pi + \frac{15}{2} \sin \pi + \frac{5\pi}{2} \right]$$

$$= \frac{1}{32} \left(-6 + \frac{15}{2} + \frac{5\pi}{2} \right)$$

$$= \frac{1}{32} \left(\frac{33}{2} + \frac{5\pi}{2} \right)$$

$$= \frac{44 + 15\pi}{192} \quad \square$$

Q6.

a(i) For $y = \cos^{-1}\left(\frac{x-2}{2}\right)$

Domain $-1 \leq \frac{x-2}{2} \leq 1$

$\therefore 0 \leq x \leq 4$

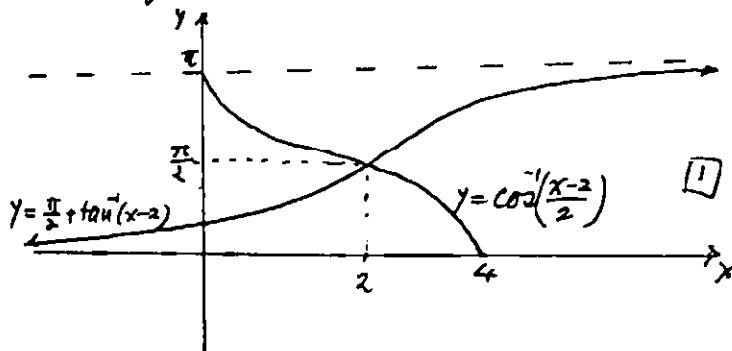
Range $0 \leq y \leq \pi$

For $y = \frac{\pi}{2} + \tan^{-1}(x-2)$

Domain: all real x

Range: $0 \leq y < \pi$

or showing clearly on the graph



(ii) Noting that $y = \cos^{-1}\left(\frac{x-2}{2}\right)$ is the translation of $y = \cos^{-1}\frac{x}{2}$ to $x=2$ while $y = \frac{\pi}{2} + \tan^{-1}(x-2)$ is obtained by translating the origin to $(2, \frac{\pi}{2})$.

$\therefore$ the intersection of $y = \cos^{-1}\left(\frac{x-2}{2}\right)$ and $y = \frac{\pi}{2} + \tan^{-1}(x-2)$ is $(2, \frac{\pi}{2})$

$\therefore$ solution to $\cos^{-1}\left(\frac{x-2}{2}\right) - \tan^{-1}(x-2) \leq \frac{\pi}{2}$

ie $\cos^{-1}\left(\frac{x-2}{2}\right) \leq \frac{\pi}{2} + \tan^{-1}(x-2)$

$\therefore x \geq 2$

(b) $\frac{\sqrt{3}+i}{\sqrt{3}-i} = \frac{(\sqrt{3}+i)(\sqrt{3}+i)}{(\sqrt{3}-i)(\sqrt{3}+i)} = \frac{2+2\sqrt{3}i}{4}$

$= \frac{1}{2} + \frac{\sqrt{3}}{2}i$

$= \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$

$\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^p = \cos \frac{p\pi}{3} + i \sin \frac{p\pi}{3}$

$\therefore$ If $\left(\frac{\sqrt{3}+i}{\sqrt{3}-i}\right)^p = \frac{1}{2}(-1 + i\sqrt{3})$

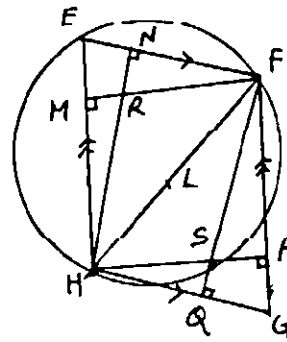
$\cos \frac{p\pi}{3} + i \sin \frac{p\pi}{3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$

ie $\cos \frac{p\pi}{3} = -\frac{1}{2}$

$\sin \frac{p\pi}{3} = \frac{\sqrt{3}}{2}$

$\therefore$ the smallest positive integer p is 2

(c)



(i) In $SPGQ$,
 $\angle P + \angle Q = 180^\circ$ ($\angle P = \angle Q = 90^\circ$, given)
 $\therefore S, P, G, Q$ are concyclic (opp $\angle$ s supplementary)
 $\therefore \angle FSP = \angle PGQ$ (ext $\angle$ of cyclic quad)
 but $\angle PGQ = \angle FEH$ (opp $\angle$ s of $\parallel$ gram)
 $\therefore \angle FSP = \angle FEH$
 $\therefore E, F, S, H$ are concyclic (ext $\angle$ equals to int opp $\angle$)
 $\therefore S$ lies on the circle.

(ii) In $HNFQ$
 $NF \parallel HQ$ (given)
 $\angle Nfq + \angle HQF = 180^\circ$ (co-int $\angle$, $NF \parallel HQ$)
 $\therefore \angle Nfq = 90^\circ$ ($\angle HQF = 90^\circ$)
 $\therefore HNFQ$ is a rectangle (all $\angle$ s 90°)
 $\therefore HR \parallel SF$ (opposite sides of rectangle)
 Similarly, $HMFQ$ is a rectangle
 $\therefore RF \parallel HS$ (opposite sides of rectangle)
 $\therefore HRFQ$ is a parallelogram.
 $\therefore HF, RS$ bisect each other (diagonal of $\parallel$ gram)
 L is mid-pt of HF (given)
 $\therefore L$ must be mid-pt of RS
 $\therefore R, L, S$ are collinear.

(iii) $\angle HMF = \angle HNF$ (rt $\angle$ s, given)
 $\therefore H, M, N, F$ lie on the semi-circle with HF as diameter.
 Similarly, F, P, Q, H is a semi-circle on HF as diameter.
 $\therefore M, N, F, P, Q, H$ is concyclic.

Q8(a)

$$(i) \left(\sqrt{\frac{x}{y}} - \sqrt{\frac{y}{x}} \right)^2 \geq 0$$

□

$$\frac{x}{y} - 2 + \frac{y}{x} \geq 0$$

$$\therefore \frac{x}{y} + \frac{y}{x} \geq 2$$

□

$$(ii) \frac{x+y+z}{w} + \frac{w+y+z}{x} + \frac{w+x+z}{y} + \frac{w+x+y}{z}$$

$$= \left(\frac{x}{w} + \frac{w}{x} \right) + \left(\frac{y}{w} + \frac{w}{y} \right) + \left(\frac{z}{w} + \frac{w}{z} \right) + \left(\frac{x}{y} + \frac{y}{x} \right) + \left(\frac{z}{y} + \frac{y}{z} \right) + \left(\frac{x}{z} + \frac{z}{x} \right)$$

□

$$\geq 2+2+2+2+2+2$$

$$\therefore \frac{x+y+z}{w} + \frac{w+y+z}{x} + \frac{w+x+z}{y} + \frac{w+x+y}{z} \geq 12$$

□

$$(iii) \frac{x+y+z}{w} = \frac{w+x+y+z-w}{w} = \frac{1-w}{w} = \frac{1}{w} - 1$$

$$\text{Similarly } \frac{w+y+z}{x} = \frac{1}{x} - 1$$

$$\frac{w+x+z}{y} = \frac{1}{y} - 1$$

$$\frac{w+x+y}{z} = \frac{1}{z} - 1$$

(2)

Substitute (2) into (1)

$$\left(\frac{1}{w} - 1 \right) + \left(\frac{1}{x} - 1 \right) + \left(\frac{1}{y} - 1 \right) + \left(\frac{1}{z} - 1 \right) \geq 12$$

□

$$\therefore \frac{1}{w} + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \geq 16$$

□

$$(b)(i) J_0 = \int_0^1 e^{-x} dx = \left[-e^{-x} \right]_0^1$$

$$= 1 - e^{-1} = 1 - \frac{1}{e}$$

□

$$(ii) J_n = \int_0^1 x^n e^{-x} dx = \int_0^1 x^n d(-e^{-x})$$

$$= \left[-x^n e^{-x} \right]_0^1 + n \int_0^1 x^{n-1} e^{-x} dx$$

□

$$= -\frac{1}{e} + n J_{n-1}$$

$$\therefore J_n = n J_{n-1} - \frac{1}{e}$$

□

$$(iii) \text{ Since } 0 \leq x^n e^{-x} \leq x^n \quad \forall 0 \leq x \leq 1$$

$$\therefore 0 \leq \int_0^1 x^n e^{-x} dx \leq \int_0^1 x^n dx$$

$$\therefore 0 \leq \lim_{n \rightarrow \infty} J_n \leq \lim_{n \rightarrow \infty} \frac{1}{n+1}$$

□

$$\text{Hence } \lim_{n \rightarrow \infty} J_n = 0$$

(iv) when $n=0$

$$J_0 = 1 - \frac{1}{e} \sum_{r=0}^{\infty} \frac{1}{r!} = 1 - \frac{1}{e} \cdot \frac{1}{0!}$$

$$= 1 - \frac{1}{e}$$

$$= J_0 \text{ from (i)}$$

$\therefore$ it is true for $n=0$

□

Assume it is true for $n=k$

$$\text{i.e. } J_k = 1 - \frac{1}{e} \sum_{r=0}^k \frac{1}{r!}$$

$$\text{then } J_{k+1} = (k+1) J_k - \frac{1}{e} \text{ from (ii)}$$

$$= (k+1) \left[1 - \frac{1}{e} \sum_{r=0}^k \frac{1}{r!} \right] - \frac{1}{e}$$

□

$$= (k+1) - \frac{(k+1)}{e} \sum_{r=0}^k \frac{1}{r!} - \frac{1}{e}$$

$$= (k+1) - \frac{(k+1)}{e} \left[\sum_{r=0}^k \frac{1}{r!} + \frac{1}{(k+1)!} \right]$$

□

$$= (k+1) - \frac{(k+1)}{e} \sum_{r=0}^{k+1} \frac{1}{r!}$$

$\therefore$ it is true for $n=k+1$ if it is true for $n=k$.

Since it is proved true for $n=0$,

$\therefore$ it will be true for $n=1, 2, 3, \dots$

i.e. true for all integers $n \geq 0$

(v)

$$J_n = 1 - \frac{1}{e} \sum_{r=0}^n \frac{1}{r!}$$

$$\therefore \frac{J_n}{n!} = 1 - \frac{1}{e} \sum_{r=0}^n \frac{1}{r!}$$

Pass the limit $n \rightarrow \infty$ on both sides

$$0 = 1 - \lim_{n \rightarrow \infty} \frac{1}{e} \sum_{r=0}^n \frac{1}{r!} \quad \text{since } J_n \rightarrow 0 \text{ as } n \rightarrow \infty \text{ by (iii)}$$

$$\therefore 1 - \lim_{n \rightarrow \infty} \frac{1}{e} \sum_{r=0}^n \frac{1}{r!} = 0$$

$$\text{or } \frac{1}{e} \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{r!} = 1$$

$$\text{i.e. } \lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{r!} = e$$

□